SMART WARDROBE

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Project Specification Document

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1. PROBLEM STATEMENT

These days, people often feel tired from making too many choices, especially when picking what to wear. It can be hard to keep track of all the clothes you own, decide which ones to wear, and match them with new items or changing weather. The Smart Wardrobe project is designed to create a smart app that helps users manage their clothes in a digital way, suggest outfits that fit their style, and check how well new items would work with what they already have.

2. PROBLEM DESCRIPTION AND MOTIVATION

Choosing daily outfits can be really stressful and take up a lot of time for many people. Often, people spend too much time deciding what to wear, forget about the clothes they already have, or buy new items that don't go well with what they already own. This leads to buying more than needed, spending extra money, and causing harm to the environment because of unnecessary purchases.

The Smart Wardrobe project is designed to fix this problem by turning the way people manage their clothes into a digital experience. Users can take pictures of their clothes, and an AI system will automatically sort and separate them. The app will then suggest outfit ideas based on things like the weather, the event they're going to, or what they like. If someone is thinking about buying new clothes, the app will help them see how those items would match with what they already have, making it easier to make better buying choices.

The motivation behind this project matters because of both social and technological reasons. Socially, it helps encourage more sustainable fashion choices and reduces the habit of buying too much. Technologically, it uses modern tools like image processing, machine learning, and mobile development platforms to create a personalized and efficient experience for users. This project is valuable because it provides a real-life solution that matches today's trends in sustainability and smart living.

3. MAIN GOAL AND OBJECTIVES

Main Goal

To design and develop an intelligent iOS-based mobile application that enables users to manage their wardrobe digitally, receive personalized outfit recommendations, and evaluate the compatibility of new clothing items using AI-powered image recognition and contextual analysis.

Objectives

Objective 1: To develop a Flutter-based iOS mobile application with user authentication and wardrobe management functionality.

Objective 2: To implement an AI-powered clothing classification and segmentation module using pre-trained deep learning models.

Objective 3: To design and develop a recommendation system that suggests outfit combinations based on user context (weather, mood, event type).

Objective 4: To enable users to check the compatibility of potential new purchases with their existing wardrobe by comparing style, color, and category.

4. RELATED WORK

4.1 Developing A Smart Wardrobe System(RFID-based)

The authors offer a system that allows users to better organize their clothing collection by taking into account color, style, event, and emotion preferences[1]. The device uses RFID technology to track garment movement within the wardrobe. While it addresses inventory management and basic recommendations, it does not focus on user-uploaded mobile images, nor on weather or mood-based outfit suggestions areas that our initiative intends to cover.

4.2 Smart wardrobe: A Comprehensive Approach to Personalized Clothing Recommendation

This work uses a multi-model method that combines rule-based classification, deep learning (ResNet-50), and K-Nearest Neighbors[2]. It analyzes uploaded clothes photographs and matches them to the user's interests and events. However, it lacks explicit integration of external context such as weather or user mood, and it is not particularly geared for iOS mobile unlike our proposed solution.

4.3 Survey Paper on Smart Wardrobe System Using Artificial Intelligence

The paper gives an overview of AI-based smart wardrobe systems, where users can upload their clothes, sort them, and get outfit ideas through web platforms. [3] It highlights the benefits of staying organized and being more sustainable, but it mainly focuses on web applications. It doesn't include a mobile-friendly design or offer more personalized suggestions based on factors like weather or mood, which our project aims to include.

4.4 Wardrobe Management Apps and Their Unintended Benefits for Fashion Sustainability and Well-Being

This research examines the role of mobile wardrobe management applications (WMAs) by analyzing user reviews. It concludes that these apps can help reduce excessive consumption and improve the use of clothing items. [4] Although it provides useful insight into user behavior and sustainability, it does not deeply explore the AI-driven combination recommendations based on image upload,nor the technical architecture for such systems.

5. SCOPE

The project focuses on integrating AI technologies for outfit recommendations, virtual try-on features, and smart wardrobe management, catering to users seeking personalized styling solutions and improved wardrobe organization.The application enables users to digitize their wardrobe by uploading photos of their clothing items and organizing them into categories such as tops, bottoms, socks and footwear.The

system provides personalized outfit recommendations based on daily weather conditions and user preferences. Each user will have a digital wardrobe that mirrors their actual clothing collection. It also aims to enhance user engagement through an intuitive interface and seamless fashion coordination.

Assumptions:

- Users should upload clear and distinguishable photos of their clothing items.
- Minor delays in AI processing or API calls are acceptable within the context of a mobile application.
- Users should have a stable internet connection when using features that require weather data or AI inference.
- Users should categorize their clothing items according to the guidance provided by the application.

Constraints:

- The project is only supported in the IOS mobile application platform.Android or web versions are not included in this project.
- The performance of on device AI models is limited by the hardware capabilities of IOS devices.
- AI-based classification results may differ depending on lighting,angle,or background circumstances in the uploaded images.
- The accuracy of outfit recommendations is dependent on the quality of the images supplied by users.

6. METHODOLOGY AND TECHNICAL APPROACH

In this section, a planned methodology for implementation of the idea “Smart Wardrobe App” will be spoken. Model Training and Mobile App Development parts included.

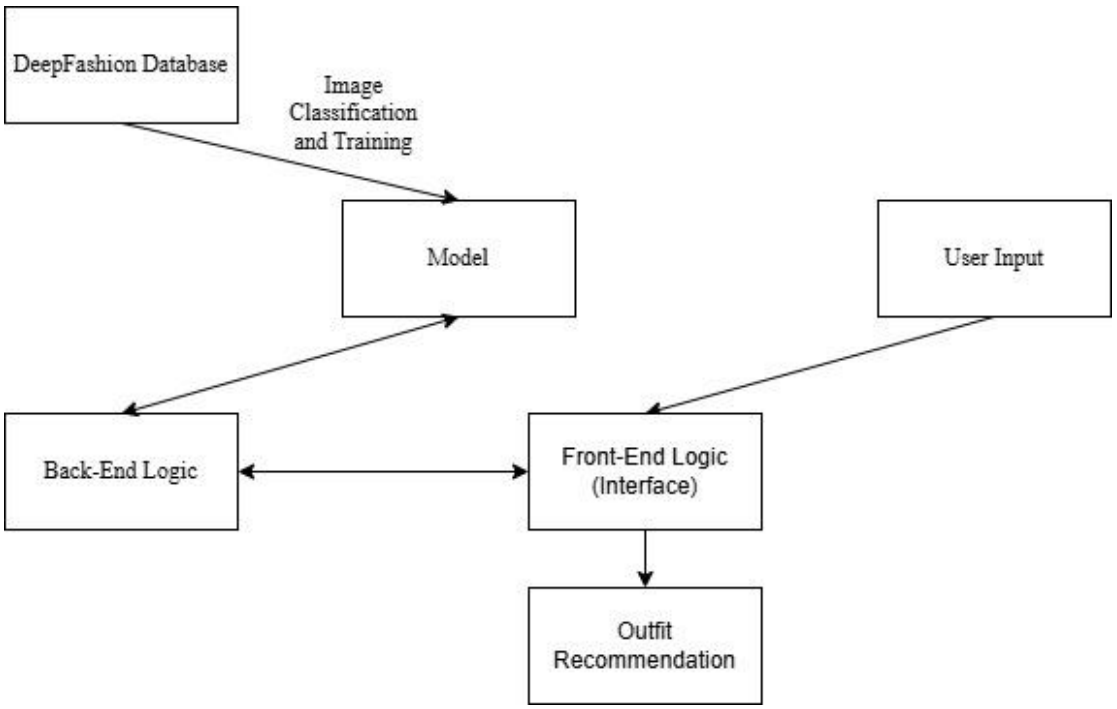

Figure 1: Structure Schema

6.1 Image Classification

6.1.1 YOLO

We will use YOLO in order to classify clothing types such as t-shirt, shirt, jacket, pants, socks, shoes, etc. With the help of YOLO's high performance classification ability, users will be able to upload a single picture for each of their clothes and get them classified regarding the type of the clothing[5].

YOLO is a pretrained model so we will be able to load YOLO by using a simple command in python.

```
from ultralytics import YOLO

model=YOLO("yolo11n.pt")
```

6.1.2 Preferred YOLO Model

The chosen YOLO model to be used is YOLOv11 considering the benefits following[6] ;

- **Improved Feature Extraction:** YOLO11 uses a better backbone and neck architecture, which improves feature extraction capabilities for complicated task performance and more accurate object detection.

- **Optimized for Speed and Efficiency:** YOLO11 has improved training pipelines and sophisticated architectural designs that provide quicker processing rates while preserving the ideal ratio of accuracy to performance.

- **Adaptability Across contexts:** To provide optimal adaptability, YOLO11 may be easily implemented across a range of contexts, including edge devices, and systems that utilize NVIDIA GPUs.

- **Wide Variety of Supported Tasks:** YOLO11 is made to handle a wide range of computer vision problems including image classification.

6.2 Fine-Tuning for Style Recognition

6.2.1 Database

As mentioned, YOLOv11 will be used for image classification but we need to find-tune and create our own model in order to make our mobile application able to work on styling recommendations. In order to achieve a good understanding of "how humans dress?". We decided to use the "Deep Fashion Database" created by "The Chinese University of Hong Kong" [7]. More than 800,000 different fashion photos, from well-posed store photos to unrestricted customer photos, may be found on DeepFashion. So this database is so useful since it includes various types of clothing style images that the images uploaded by the users will have similarities.

6.2.2 Fine-Tuning Principles

We will implement a single ResNet backbone with two heads which predicts color and pattern in one pass[8]. The system will be able to quickly locate visually related items in the user's wardrobe thanks to a fast vector index. The system will use explainable stylistic rules (patterned tops match best with solid bottoms, for example) in addition to basic similarity. As it develops, it will also learn a "compatibility" model that identifies components that typically work well together in practical applications.

To get the network to quickly learn our categories and properties, we will start by training the model's heads. The backbone is then progressively unfrozen so that it can adapt to the unique distribution and appearance of fashion photos without becoming overfit. After training the recognition model, we will generate embeddings by running each catalog image through the backbone. Because these vectors are kept in a fast index[9], visually related objects can be instantly retrieved during recommendation.

6.3 Reasoning for Different Conditions

6.3.1 Context Integration

A basic context object will be used to express occasion, mood, and weather. The occasion dictates the dress code, the weather imposes "hard" restrictions to guarantee comfort, safety and the mood adjusts the intensity of the pattern and palette. These inputs change ranking scores and remove inappropriate items if necessary.

6.3.2 Recognition Results

The model will generate a visual embedding and predict the color, and pattern of the item. There will be a confidence score associated with each forecast[10].

6.4 Recommendation Logic

6.4.1 Selection Algorithm

Candidates will be selected in the user's wardrobe (to mix and match their own goods). Images, embeddings, and basic metadata (category, color, pattern, materials, and care aspects) are used to describe each contender.

6.4.2 Weather

Strong restrictions are added by the weather. The method will promote lighter, breathable options in hot weather; it will make sure that there is enough layering in cold weather; it will favor water-resistant options in rainy weather; and it will steer clear of pieces that are too flowing in winds. Before rating, items that are heavily inappropriate considering weather requirements will be filtered out.

6.4.3 Occassion

Occasion corresponds to a desired degree of formality and usefulness. The system will prefer clean shapes and simple patterns for an office situation; it will combine comfort, mobility, and long-lasting footwear for a concert; and it will lean toward polished, coordinated ensembles for a date. Which candidates are ranked higher or lower depends on the occasion.

6.4.4 Mood

Color scheme and pattern intensity are influenced by mood. While worn-out moods tend toward comfort-first cuts and neutral, soothing tones, cheerful moods can call for bolder pattern choices or brighter accents. Quiet emotions prioritize muted palettes and simpler patterns. Mood rarely filters items directly. Instead, it guides the ranking in the direction of those that "feel right" considering users' mood preference at the time.

6.5 Storing Users' Clothes

We will save the original image, a standard thumbnail, the embedding, and metadata (generic tags like fabric qualities, category, color, and pattern) for every item. This makes it possible to retrieve information quickly, provide clear explanations, and re-score objects when models or rules advance.

6.6 Back-End Logic

A basic collection of authorized endpoints that manage wardrobe/catalog maintenance, suggestions, and recognition will be exposed by the backend. The service will save images in object storage and will save metadata in a relational database, and uses a ResNet-based recognizer to retrieve the garment category, color, pattern, and visual embedding when the mobile app uploads an image or references an existing item.

The recommendation pipeline will use the vector index to find candidates in the user's wardrobe, will apply hard filters for safety and constraints (like stock/size rules and weather compatibility), and then will calculate a final ranking that combines context scoring (mood, weather, occasion), explainable style rules (pattern balance,

color harmony, and appropriate formality/fabric), and visual similarity. The results are returned as entire outfits. Background workers will manage asynchronous embedding extraction, index maintenance, catalog ingestion, and model rollouts; and standard security procedures (JWT, signed media URLs) safeguard the API and user information[11].

6.7 Front-End Logic

After guiding the user via a straightforward capture-or-upload process, the mobile frontend will use rapid selections to gather context (mood, occasion, and weather). Once an image has been selected, the application will send the image to the backend /recognize endpoint. With editable chips, the user interface will show the identified category, color, and pattern so that errors can be fixed. A thumbnail of the item will be added to "My Wardrobe.". In order to make recommendations, the app will compile the chosen item along with context (such as occasion, mood, and weather), call /recommend/outfit, and then produce entire ensembles. The user interface will gracefully manage failures and loading. Performance and accessibility will be integrated to ensure a seamless experience across devices and network conditions.

7. PROFESSIONAL CONSIDERATIONS

7.1 Methodological Considerations/Engineering Standards

- **Google Colab:** We will use the T4 GPU provided by Google Colab in order to train and test our model. T4 is a high-performance GPU produced by NVIDIA.
- **Visual Studio Code:** We will use Visual Studio Code and its libraries to build our application's back-end side and front-end side.
- **Python:** The model inference pipeline, preprocessing, recommendation logic, backend services, background jobs, and integrations will all be put into place.
- **Pytorch and torchvision:** We will construct embeddings, execute inference in the recognition service, and fine-tune a ResNet backbone with three heads (category, color, and pattern).

- **Numpy and scikit-image:** Before and after model inference, we will carry out image operations, format conversions, and light preprocessing.
- **FAISS:** To begin making outfit recommendations, we will index the embeddings of catalog and wardrobe goods and get the best candidates.
- **FastAPI for REST APIs:** When necessary, we will incorporate background tasks, validate requests, provide async I/O, and expose endpoints like /recognize and /recommend/outfit.
- **PostgreSQL:** Users, items, context (mood/occasion), embedding references, metadata (category/color/pattern), and audit logs will all be stored.
- **Flutter:** The mobile application will be developed to take and upload photos, gather moods and occasions, make calls to backend APIs, and show costumes and recognition.
- **JWT and Signed URLs:** JWT will be used to authenticate API requests, and signed URLs will be used for safe, direct picture uploads and downloads to object storage without the need for an API proxy.
- **Postman:** We'll set up environments, generate collections for key endpoints, do automated regression tests, and distribute examples across team members.

7.2 Realistic Constraints

7.2.1 Economic

Since all of the tools we will use are free to use, we do not expect any costs during the development of our mobile app project. Different costs may occur if we desire to publish and make our app available on mobile app stores. Usage of Google Colab will be cost-efficient and there will be no need to consider GPU upgrading expenses. An important aspect to mention is this application will help users to save money since they will be able to style their outfits with the items they already have in their wardrobe rather than buying new items in order to create new outfit variations for themselves.

7.2.2 Environmental

There are no environmental concerns.

7.2.3 Ethical

The biggest ethical concern is about the privacy of the images. We will consider security measures throughout the development process in order to prevent any ethical issues regarding safety of individuals' data privacy and there will not be any public access to other's wardrobe which includes images uploaded to the application's wardrobe section.

7.2.4 Health and Safety

To consider psychological well-being, there will not be any judgement of the size of the clothes nor clothing style of the individuals. Other than psychological concerns, our app is not harmful in any aspect of human health.

7.2.5 Sustainability

Since trends in fashion change massively over time, we are planning to support our application by continuous software updates in order to extend its lifespan by improving the recommendation logic progressively and updating our database considering new trends.

7.2.6 Social

The recommendation algorithm will not be based on any specific sociological groups' clothing preference. Members of any race or any sociological group will not experience discrimination regarding outfit recommendations. Instead, they will experience various kinds of outfit recommendations that they never considered to wear in their social lives. Meaning this app will provide a variety of options for their specific clothing needs in their daily lives.

7.3 Legal Constraints

7.3.1 Database

The DeepFashion database is prepared by The Chinese University of Hong Kong. The official website of the database clearly states legal constraints for whom that uses the database. We find it useful to include the “Agreement” section they put on their website here. As follows:

Agreement

- The DeepFashion is available for **non-commercial** research purposes only.
- All images of the DeepFashion are obtained from the Internet which are not property of MMLAB, The Chinese University of Hong Kong. The MMLAB is not responsible for the content nor the meaning of these images.
- You agree **not to** reproduce, duplicate, copy, sell, trade, resell or exploit for any commercial purposes, any portion of the images and any portion of derived data.
- You agree **not to** further copy, publish or distribute any portion of the DeepFashion. Except, for internal use at a single site within the same organization it is allowed to make copies of the dataset.
- The MMLAB reserves the right to terminate your access to the DeepFashion at any time.

We agree that there will be **no** publication of our application’s version that uses DeepFashion database on mobile app store platforms.

7.3.2 Processing Images and Sensitive Data Risks

There will be no data processing for facial data or precise geolocation included in the images for any purpose of outfit recommendations.

7.3.3 Consent

Any image processing requires the consent and permission of the user. By uploading clothes to the application, the user is considered as they have stated their consent for image processing.

8. MANAGEMENT PLAN

8.1 Phases

- **Phase 1: Literature Survey (1 month)**

To establish baselines, risks, and ethical/licensing limitations, we will examine previous research, datasets, and models.

- **Phase 2: Data Preparation (2 weeks)**

To guarantee that the model learns effectively, feature extraction and preprocessing procedures will be carried out due to the amount of the data. This stage will assist in improving the data's manageability and significance for the educational process.

- **Phase 3: Developing Image Classification and Reasoning Algorithm (1 month)**

YOLO11 will be used. We will fine-tune a ResNet, export a model that is suitable for inference, and calibrate confidence.

- **Phase 4: Creating Logic Regarding Mood, Occasion and Weather Aspects (2 weeks)**

We will establish a context schema and set of rules, such as weather as strict limitations, occasion as a dress code, and mood as subtle styling cues.

- **Phase 5: Developing Outfit Recommendation Algorithm (5 weeks)**

In addition to introducing diversity and useful filters, we will create a vector index, retrieve candidates, and rank them using similarity + rules + context.

- **Phase 6: Developing Back-End Logic (7-8 weeks)**

FastAPI services, storage, and background jobs will be implemented; JWT and signed URLs will be used for security; and observability will be added.

- **Phase 7: Developing Front-End Logic (7-8 weeks)**

In order to capture inputs and correctly handle mistakes, we will construct a Flutter application.

- **Phase 8: Final Adjustments and Documentation (1 month)**

We will complete papers, optimize performance, correct issues, and package a non-commercial demo.

8.2 Division of Responsibilities

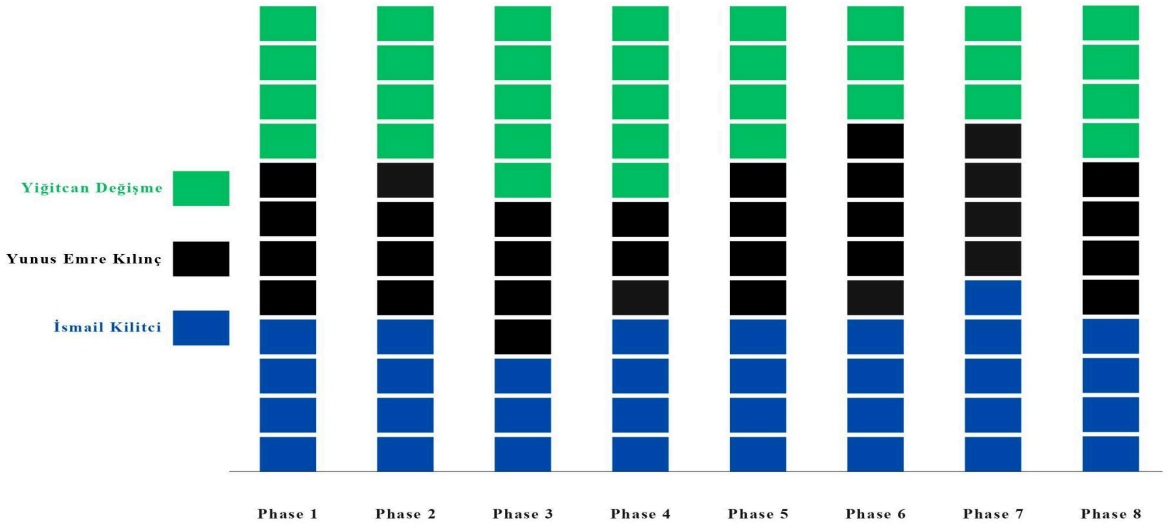

Figure 2: Distribution of Work

8.3 Timeline

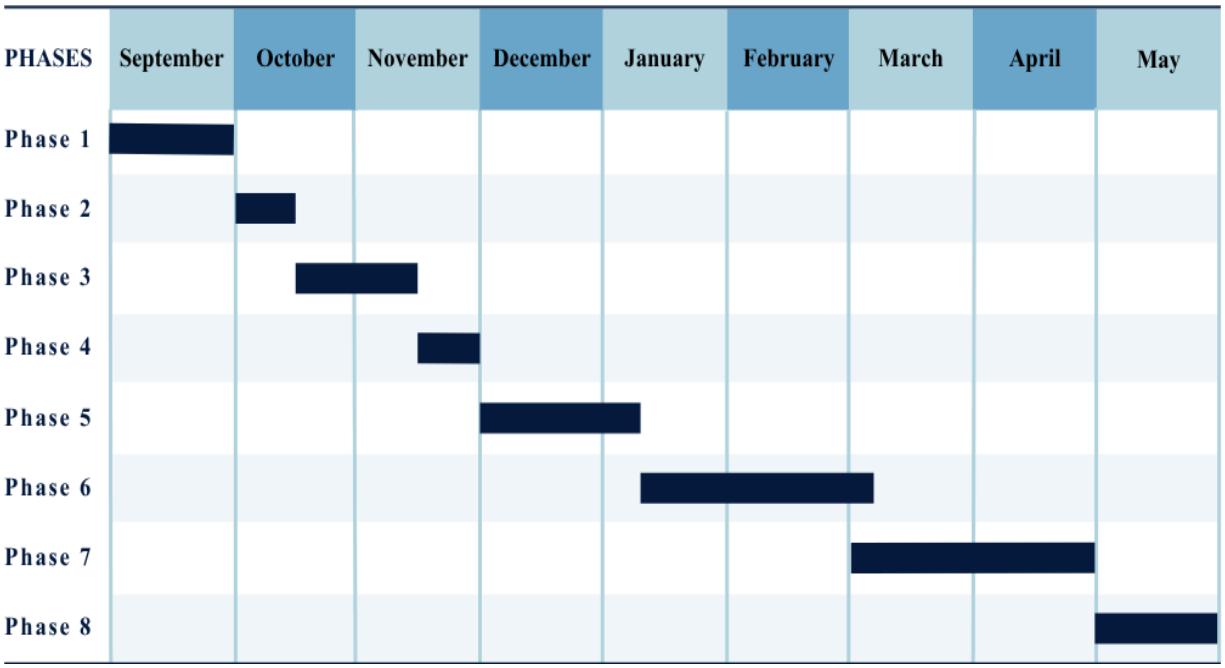

Figure 3: Project Timeline

9. SUCCESS FACTORS AND RISK MANAGEMENT

9.1 Measurability/Measuring Success

In order to evaluate the success of the Smart Wardrobe project, each objective defined in Section 3 is associated with measurable success factors and corresponding key performance indicators (KPIs).

Objective 1: To develop a Flutter-based iOS mobile application with user authentication and wardrobe management features.
Success Factor: The mobile application must allow users to successfully register, log in, and upload at least 50 clothing images to their digital wardrobe without crashes or major UI bugs. Usability tests with at least 10 participants should confirm that all core functions (authentication, upload, deletion, and viewing) work properly with at least 90% task-completion rate.

Objective 2: To implement an AI-powered clothing classification and segmentation module using pre-trained deep learning models.
Success Factor: The AI model should achieve a minimum of 85% classification

accuracy and correctly segment clothing regions in at least 80% of uploaded images, evaluated using Intersection over Union (IoU) and F1-Score metrics on a labeled test dataset.

Objective 3: To design and develop a recommendation system that generates outfit suggestions based on user context (weather, mood, or event type).
Success Factor: The recommendation module should produce contextually appropriate and stylistically coherent outfit suggestions for at least 90% of test scenarios, validated through user satisfaction surveys where at least 80% of participants rate the recommendations as “relevant” or “very relevant.”

Objective 4: To enable users to check compatibility between potential new purchases and their existing wardrobe items.
Success Factor: The compatibility check should yield results consistent with human judgment in at least 85% of cases, measured through comparison with expert-labeled compatibility ratings. The feature should deliver a compatibility score and textual feedback within 3 seconds of input.

Overall project success will be determined if all the above quantitative and qualitative targets are achieved, ensuring that the system performs accurately, efficiently, and provides a satisfactory user experience.

9.2 Risk Management

Here is a list of potential risks, the associated phases of the project, and resolution strategies (B-plans) designed to address them.

Risk-1: AI model may produce inaccurate outfit recommendations due to insufficient or biased training data.

Resolution(B-plan):Expand the dataset by incorporating publicly available fashion datasets and user-generated content. Regularly fine-tune the model with diverse samples to improve recommendation accuracy.

Risk-2: Integration issues between the AI module and the iOS interface

Resolution(B-plan): Employ modular development practices and conduct integration testing at each stage. Maintain clear communication between frontend and backend teams to ensure API compatibility.

Risk-3: Limited dataset diversity causing bias in outfit recommendations

Resolution(B-plan): Incorporate publicly available fashion datasets with different body types, styles, and cultures. Allow users to provide feedback on recommendations to help improve model fairness and accuracy.

10. BENEFITS AND IMPACT OF THE PROJECT

The proposed smart wardrobe application provides several user-centered and technological benefits. By allowing users to upload and categorize their clothing items, it enhances organization and reduces the time spent deciding what to wear. The integration of artificial intelligence enables the system to recommend outfit combinations based on daily weather conditions and the user's mood, improving personalization and user satisfaction.[12] Additionally, by helping users visualize and utilize their existing clothes more efficiently, the application encourages sustainable fashion habits and reduces unnecessary clothing purchases. This not only supports individual cost savings but also promotes environmentally conscious consumption behaviors.

10.1 Scientific Impact

This project contributes to the growing field of artificial intelligence applications in fashion technology by integrating computer vision, machine learning, and user-behavior analysis within a single IOS platform.By including contextual elements like weather and mood, which are frequently disregarded in conventional models, it promotes research on personalized recommendation systems. The system's architecture also facilitates further research on user experience optimization, human–AI interaction, and sustainable fashion behavior. The project establishes a scientific basis for future multidisciplinary research in smart systems and computational design by connecting AI and digital fashion management.

10.2 Economic/Commercial/Social Impact

The smart wardrobe application has the potential to impact retail dynamics and consumer behavior from an economic and commercial perspective. Through tailored suggestions, it can assist consumers in making more thoughtful purchases, which will cut down on unnecessary spending while also opening doors for partnerships with e-commerce sites and fashion labels. Socially, it raises awareness of excessive consumerism in the fashion industry and supports sustainable consumption. Along with helping users manage their lifestyle more effectively and gain confidence in their wardrobe choices, the app also promotes self-expression and better well-being in day-to-day interactions.

10.3 Potential Impact on New Projects

Future advancements in personalized recommendation systems and smart living may be built upon the technological framework created in this project. Other domains, like digital style advisors, smart shopping assistants, or virtual fitting rooms, can benefit from the integration of AI, image recognition, and contextual data. Additionally, the dataset and algorithms created here may serve as the basis for future initiatives centered on wearable technology integration, user behavior modeling, and sustainability analytics. In this approach, the project not only provides a useful product but also creates opportunities for further study and business use of AI-powered lifestyle management systems.

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